Kartikeya Gangwar

Computational Mathematics | Scientific Machine Learning | Geometric Deep Learning
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Education

University of Delhi **Delhi, India**
B.Sc. (Honours) Mathematics (4-Year Undergraduate Research Degree) *2023 – 2027*

- **Undergraduate Dissertation Supervisor:** Prof. Vinay Kumar (Department of Mathematics).
- Focus on real analysis, differential geometry, numerical PDEs, and building custom PyTorch/C++ autodiff routines alongside the formal degree coursework.

Technical Working Knowledge

Methods & Theory: Symplectic & contact manifolds, differential geometry (Sobolev $W^{1,2}$, Riemannian curvature, Weingarten map), Ambrosio–Tortorelli variational phase fields, Navier–Stokes formulations, free-boundary obstacle PDEs, and optimal transport.
Scientific Machine Learning & Vision: Multi-task vision transformers (PRISM, DINOv2/v3), Mallat wavelet MRA, inverse rendering optics (Spherical Harmonics, Snell-Descartes specular reflection), Contact-SSMs, Hamiltonian NNs, and parameter subspace AMR.
Computing & Systems: PyTorch (`torch.func.vmap`, forward-mode JVP/VJP, Hessian trace contractions, PCGrad/CAGrad), C++, Lua, Linux/Bash, Git, and CUDA/GPU profiling.

Research Manuscripts & Preprints

Shape Inversion in Electrical Impedance Tomography | *IEEE Trans. Comput. Imaging* **Under Review**

- **Kartikeya Gangwar** (Sole Author) **[MS #IP-106023]** Code: eit-neural-surrogate-inversion DOI: 10.5281/zenodo.22096368
- Parameterized closed conductivity boundaries with 64D periodic cubic B-splines, matching FEM boundary sensitivities against directional finite differences ($< 3.2 \times 10^{-4}$).
- Supervised directional JVPs randomly sampled over $\mathbb{S}^{63}$. This collapsed the autograd backward graph from 64 tangent passes to a single branch, keeping peak VRAM under 343 MB and yielding a 56.8× speedup (35.6 ms vs 2.02 s/step) with zero online FEM calls.

Contact-SSM: Geometric State-Space Models for Time Series | *Target: NeurIPS / SIMODS* **Working Paper**

- **Kartikeya Gangwar** (Lead Investigator) Code: contact-ssm
- Mapped latent dynamics to an odd-dimensional contact manifold $\mathbb{R}^{2d+1}$ with contact 1-form $\alpha = \mathrm{d}s - \mathbf{p}^T \mathrm{d}\mathbf{q}$ and Reeb vector field $R = \partial/\partial s$, modeling dissipative and driven systems without artificial energy decay.
- Structured the infinitesimal generator $K \in \mathfrak{sp}(2d) \oplus \mathbb{R}$ via Cayley transforms and integrated it directly into GPU associative prefix scans, achieving $\mathcal{O}(L)$ parallel training and zero-latency $\mathcal{O}(1)$ streaming rollouts.

CHRONOS: Symplectic Potential Inversion in Astrodynamics | *Target: MNRAS / A&C* **Dissertation**

- **Kartikeya Gangwar** (Lead Author), **Prof. Vinay Kumar** (Supervisor) Code: chronos-astrodynamics
- Decoupled rotating-frame kinetic terms from unknown scalar gravitational potentials. Enforced harmonic vacuum conditions ($\nabla^2 \mathcal{V} = 0$) to eliminate spurious deep-space potential wells.
- Replaced global Fourier features with compact Ricker wavelets to stop far-field ringing. Recovered all 5 Lagrange points (L_1–L_5) with exact stability matches and reduced Jacobi drift by 412× via adaptive Sundman time-dilation.

Operator Conditioning & False Convergence in Incompressible PINNs | *Physics of Fluids* **Under Review**

- **Kartikeya Gangwar** (Sole Author) **[MS #POF26-AR-14856]** Code: pinn-fluid-formulations DOI: 10.5281/zenodo.22979276
- Showed why mesh-free PINNs fail on 2D regularized cavity flow ($Re = 1000$) when written in streamfunction-vorticity (ψ–ω): without discrete Thom wall stencils, boundary vorticity drifts, causing false convergence (67× residual drop while underpredicting secondary corner eddies by 14%).
- Proved that retaining pressure p in a continuous ψ–p formulation implicitly enforces Helmholtz-Hodge projection, dropping global velocity error from 14.2% to 2.8%.

Extreme-Re Navier–Stokes Framework up to $Re = 100,000$ | *IJNMHFF (Emerald)* **Under Review**

- **Kartikeya Gangwar** (Sole Author) **[MS #HFF-10-2026-1582]** Code: lid-driven-cavity-cfd DOI: 10.5281/zenodo.18312938
- Wrote an exact $\mathcal{O}(N^2 \log N)$ Discrete Sine Transform (DST-I) Poisson solver, eliminating iterative pressure drift. Computed steady continuation to $Re = 100,000$ on 1025^2 grids with zero artificial viscosity up to $Re = 50,000$ (Richardson GCI 0.48%). Resolved periodic shedding ($St = 0.6249$) on 513^2 meshes.

The Adaptive Subspace (AS) Optimization Lineage

Act I: Parameter Decoupling & Interface Transmission in PINNs | *Neural Networks (Elsevier)* **Under Review**

- **Kartikeya Gangwar** (Lead & Corr.), **Samarjeet Malik** **[MS #NEUNET-D-26-09502]** Code: null-space-pinn DOI: 10.5281/zenodo.22132799
- Split network parameter manifolds into orthogonal direct-sum subspaces ($\Theta_0 \oplus \Theta_1$) coupled via C^2 Quintic Hermite routing. Proved structural gradient orthogonality ($\langle \nabla \mathcal{L}_{\text{if}}, \nabla \mathcal{L}_{\text{des}} \rangle \equiv 0$) on saturated domains and tested on 3D Paul ion traps.

Act II: Adaptive N-Subspace PINN (AS-PINN) | *Target: SISC* **Working Paper**

- **Kartikeya Gangwar** (Sole Author) Code: as-pinn DOI: 10.5281/zenodo.22822521
- Tracked per-sample Gram conflicts via `torch.func.vmap` to trigger parameter-space cleavage invariance ($\|u^{(N+1)} - u^{(N)}\| = 0$). Yielded 725× lower error on high-frequency Helmholtz ($k = 4\pi$) compared to PCGrad and CAGrad.

Act III: PRISM: Partition-of-Unity Multi-Task Vision Architecture | *Target: IEEE TPAMI / CVPR* **Working Paper**

- **Kartikeya Gangwar** (Sole Author) Code: prism-multitask-vision DOI: 10.5281/zenodo.22974130
- Formulated autonomous parameter-space AMR for multi-task vision backbones, continuously tracking inter-task Gram clashes via `torch.func.vmap` with function-preserving cleavage and C^2 Quintic Hermite transition gating.
- Integrated Haar 2D Wavelet MRA ($V_0 \oplus W_0$), Sobolev $W^{1,2}$ tangent orthogonality, and closed-loop inverse rendering (order-2 Spherical Harmonics, Snell-Descartes specular reflection, ray-marched shadows, and recurrent photometric self-correction $|I - \hat{I}| \to \Delta Z, \Delta \mathbf{n}$).
- Engineered Ambrosio–Tortorelli variational phase-field relaxation ($s \in [0, 1]$) for sharp occlusion cliffs with Projective Hyperbolic Manifolds ($\partial Z / \partial u = -\frac{Z^2}{d} \frac{n_x}{f_x}$); benchmarked across Meta DINOv2 and DINOv3 backbones on authentic NYUv2 raw Kinect RGB-D.

Stochastic PDEs & Free-Boundary Inversion

High-Dimensional American Basket Options (Deep-EEP-PINN) | *Target: Quantitative Finance* **Working Paper**

- **Kartikeya Gangwar** (Sole Author) Code: Deep-EEP-PINN
- Priced American basket options up to $d = 50$ assets (1,225 correlations) by combining analytical European base pricing with a boundary-encoded early-exercise premium. Evaluated multi-asset cross-diffusion via directional Hessian trace contractions in $\mathcal{O}(d)$ complexity (< 3 GB VRAM; 0.37% error vs 100k-path Longstaff-Schwartz Monte Carlo).

Bayesian PINN Posterior Fidelity & Error Localization | *Target: Inverse Problems* **Working Paper**

- **Kartikeya Gangwar** (Sole Author) Code: PINN-Bayesian-Posterior-Fidelity DOI: 10.5281/zenodo.23227155
- Formulated the Bayesian Fidelity Ratio (BFR) using 1-Wasserstein optimal transport normalized by paired MCMC stochastic noise floors to measure forward PINN surrogate error propagation into inverse posteriors.

Systems Engineering & Academic Service

BeamNG Research SDK (BRSDK v2.0.1) | *Lua / Python / Systems* **Open-Source Framework**

- **Kartikeya Gangwar** (Sole Architect) Code: BRSDK DOI: 10.5281/zenodo.21729606
- Engineered in-engine telemetry extraction inside BeamNG's 2,000 Hz soft-body physics loop with zero dynamic Lua heap allocations during logging. Built columnar Apache Arrow/Polars Python bindings and zero-copy DLPack PyTorch wrappers.

Verified Peer Reviewer | *Physics of Fluids (AIP Publishing)* **2026**

- Invited referee evaluating Large-Eddy Simulation (LES) vorticity transport and modal dynamics in computational fluid dynamics. Officially verified by publisher via ORCID (ORCID: 0009-0009-1973-7532).